

# External Adversarial Audit -- Paper II (exact fractional-cover theorem)

|                  |                                                                                                |
|------------------|------------------------------------------------------------------------------------------------|
| Audit ID         | PAPER-II-EXT-AUDIT-001                                                                         |
| Date             | 2026-07-10                                                                                     |
| Target           | max over chordal G ( $ E -2 \tau_3(G)$ ) = floor( $((2n+1)^2/24)$ ), attained by $S_{\{p,q\}}$ |
| Environment      | Python 3.14.4 / scipy 1.17.1 HiGHS / networkx 3.6.1; Lean v4.28.0 / lake 5.0.0                 |
| Master seed      | 20260710                                                                                       |
| EXECUTIVE STATUS | PASS                                                                                           |

## Executive Verdict -- PASS

Every gate of the specification was checked and none failed. The three distinct claim-types are kept separate and each is supported on its own terms: (i) the ANALYTIC PROOF is internally correct -- Lemmas 3.1 and 5.1, the C4 necessity, the complete-split closed forms (Prop 6.2/Cor 6.3), and the integer maximization (Prop 7.1) were independently re-derived/reproduced with the auditor's own code and matched the manuscript; (ii) the LEAN CERTIFICATE holds -- PaperII.theorem\_1\_2 re-elaborated (lake build, 8057 jobs, success), hypotheses exactly the standard IsChordal, content = upper bound AND attainment; (iii) the COMPUTATION found no counterexample -- exhaustive over all labelled chordal graphs  $n \leq 6$ , max = floor( $((2n+1)^2/24)$ ), attained by a complete-split, 0 violations.

The only honest residual is on INDEPENDENCE of the Lean build (Mathlib came from the project's pinned cache, not a clean-room from-scratch rebuild) and on the commit-5f2d448 SHA match, which could not be verified here (no git history reachable; the release package does not bundle the lean/ tree). These are provenance caveats, not mathematical defects. The manuscript does NOT over-claim: cp(G) and Erdos #81 are explicitly out of scope.

## Gate table (A-J)

| Gate                           | Status                      | Basis                                    |
|--------------------------------|-----------------------------|------------------------------------------|
| A Definitions                  | VERIFIED                    | consistent manuscript/ledger/Lean        |
| B Lemma 3.1 vertex-copy        | VERIFIED (in range)         | P2: 11032 pairs, 0 violations            |
| C Lemma 4.1-4.3 + C4 necessity | VERIFIED                    | P5: C4 necessity confirmed               |
| D Lemma 5.1 terminal char.     | VERIFIED                    | P5: exhaustive $n \leq 6$ , 0 counterex. |
| E Sec6 complete-split cover    | VERIFIED                    | P3: closed forms == full LP              |
| F Sec7 integer maximization    | VERIFIED                    | P4: exact $n \leq 5000$                  |
| G Global assembly (Thm 1.1)    | VERIFIED                    | P1: exhaustive $n \leq 6$ max=M(n)       |
| H Formal certificate           | VERIFIED_WITH_RESIDUAL_RISK | build OK; axioms; SHA/indep. residual    |
| I Citations                    | VERIFIED                    | A1-A6 standard, in-scope                 |
| J Over-claim / scope           | VERIFIED (no over-claim)    | cp/#81 explicitly out of scope           |

## Computational tests (P1-P5) + Lean (H)

| Test             | Gate | Coverage                                             | Verdict                         |
|------------------|------|------------------------------------------------------|---------------------------------|
| P1 max Phi_tau   | F,G  | exhaustive $n \leq 6$ + CS scan $n \leq 60$ + random | VERIFIED / no counterexample    |
| P2 vertex-copy   | B    | exhaustive $n \leq 5$ + random (11032 pairs)         | VERIFIED                        |
| P3 closed forms  | E    | full cover-LP vs closed form, (p,q) 0..12            | VERIFIED                        |
| P4 integer max   | F    | exact Fraction, $n=1..5000$                          | VERIFIED                        |
| P5 terminal + C4 | C,D  | exhaustive chordal $n \leq 6$                        | VERIFIED                        |
| H Lean           | H    | lake build (8057 jobs) + #print axioms               | build OK; residual on indep/SHA |

## Gate H -- Lean axiom footprint (#print axioms PaperII.theorem\_1\_2)

Build: 'Build completed successfully (8057 jobs)'. Statement hypotheses exactly IsChordal (Unconditional.lean:72-79); IsChordal is the standard chord definition (every cycle of length  $\geq 4$  has a chord, Chordal.lean:35). Captured axioms:

```
'PaperII.theorem_1_2' depends on axioms: [propext, Classical.choice, Quot.sound]
theorem_1_2 :  $\forall (n : \mathbb{N}),$ 
   $1 \leq n \rightarrow$ 
    ( $\forall \{W : \text{Type}\} [\text{inst} : \text{Fintype } W] [\text{inst}_1 : \text{DecidableEq } W] (G : \text{SimpleGraph } W) [\text{inst}_2 : \text{DecidableRel } G.\text{Adj}],$ 
       $\text{Fintype.card } W = n \rightarrow \text{IsChordal } G \rightarrow \text{phiTau } G \leq \uparrow((2 * \uparrow n + 1) ^ 2 / 24)) \wedge$ 
       $\exists W \times x_1 G x_2, \text{Fintype.card } W = n \wedge \text{IsChordal } G \wedge \text{phiTau } G = \uparrow((2 * \uparrow n + 1) ^ 2 / 24)$ 
    )
@IsChordal :  $\{V : \text{Type } u_1\} \rightarrow \text{SimpleGraph } V \rightarrow \text{Prop}$ 
def PaperII.IsChordal. $\{u_1\} : \{V : \text{Type } u_1\} \rightarrow \text{SimpleGraph } V \rightarrow \text{Prop} :=$ 
fun  $\{V\} G =>$ 
   $\forall v : V (c : G.\text{Walk } v v),$ 
     $c.\text{IsCycle} \rightarrow 4 \leq c.\text{length} \rightarrow \exists x \ y, x \in c.\text{support} \wedge y \in c.\text{support} \wedge G.\text{Adj } x \ y \wedge s(x, y) \notin c.\text{edges}$ 
```

## Falsification log (all NEGATIVE)

No chordal  $G$  with  $\text{Phi\_tau} > \text{floor}((2n+1)^2/24)$  (exhaustive  $n \leq 6$ : 18154 chordal graphs at  $n=6$ ; CS scan  $n \leq 60$ ; random chordal  $n=7..14$ ). No non-complete-split maximiser. No vertex-copy failure in 11032 pairs. No statement divergence manuscript/ledger/Lean. A bounded search is supporting evidence, not proof; the  $n \leq 6$  enumeration is exhaustive hence conclusive for those  $n$ .

## Coverage & residual risk

Checked: all definitions; Lemmas 3.1/4.3/5.1; Sec6 closed forms; Sec7 integer max (exact to  $n=5000$ ); global exhaustive max  $n \leq 6$ ; Lean build + statement shape + axioms; citations; scope. Not checked / residual: clean-room Mathlib rebuild; commit-5f2d448 SHA match (git unreachable here); exhaustive chordal  $n \geq 7$  (infeasible; covered by sampling + analytic proof + Lean); Spanish\_es manuscript beyond spot check.

## Final recommendation

PASS. No mathematical objection to the boxed theorem; analytic proof, formal certificate, and exhaustive small-case computation agree. Non-blocking closures: (1) reproduce lake build + #print axioms at commit 5f2d448 in a clean-room environment with a from-scratch Mathlib; (2) publish the lean/ tree with a SHA-256 manifest inside the release package. This internal adversarial audit confers NO peer-review or publication status.